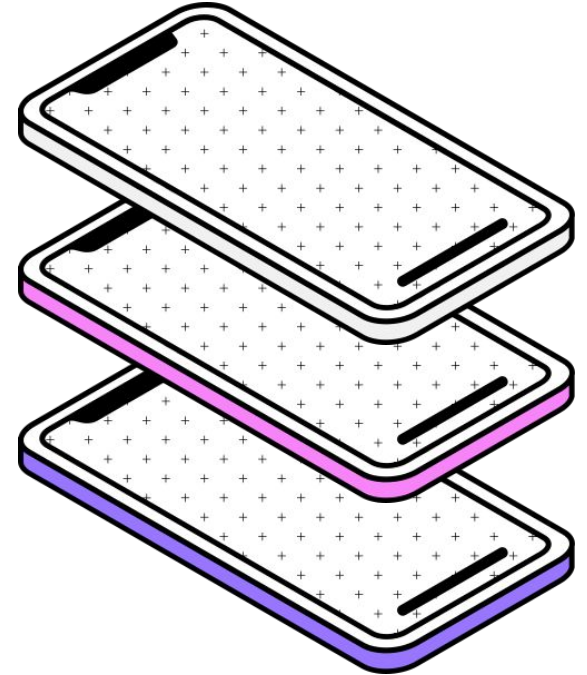


Mobile Security Solutions @ C-DAC



Presented by
M- Sec Team

M-Kavach AppSec SDK



इलेक्ट्रॉनिक्स एवं
सूचना प्रौद्योगिकी विभाग
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY





M-Kavach AppSec SDK

Mobile applications are increasingly exposed to risks because of inadequate real-time protection. The M-Kavach AppSec SDK addresses these challenges by monitoring the secure posture of the device at runtime, identifying potential threats, and enforcing robust security. This enables developers to seamlessly integrate the SDK to deliver secure and compliant mobile applications efficiently, without impacting development timelines.

Salient Features

 Root Detection Detects rooted and compromised devices	 Emulator Detection Detects if application is running on emulated/virtualized environment
 Debug Detection Detects if USB/Wireless debugging are enabled & any active debugger attached	 Device & Network Security Checks Perform various security checks such as VPN, Proxy, SecureWiFi, PlayProtect, LockScreen etc
 Instrumentation Tools Detection Detects the presence of instrumentation tools	 Screen Capture Prevention Prevents capturing of the application screens

M-Kavach AppSec SDK Architecture



With M-Kavach AppSec SDK

```

graph TD
    A[App running on Compromised Device] --> B[Misconfigurations Detected  
M-Kavach SDK scans and identifies vulnerabilities]
    B --> C[Application Response  
Allow Functionality / Block / Restrict Functionality]
    C --> D[Data is Safe]
            
```



Without M-Kavach AppSec SDK

```

graph TD
    A[App running on Compromised Device] --> B[Misconfigurations Not Detected]
    B --> C[Application Response  
Application Allows Functionality]
    C --> D[Data is Compromised]
            
```

Why M-Kavach AppSec SDK?



Works across diverse Android & iOS devices



Supports offline and low-connectivity environments



Designed for large-scale deployments

Target Beneficiaries



BANKING & FINANCIAL SERVICES



GOVERNMENT APPLICATIONS



ENTERPRISE MOBILITY SOLUTIONS



LARGE-SCALE SURVEY & FIELD APPLICATIONS

Contact Details :

✉ mkavach@cdac.in

☎ +91 90142 02885

प्रगत संगणन विकास केंद्र
CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

M-Kavach AppSec SDK

- Mobile applications are increasingly exposed to risks because of inadequate real-time protection.
- The M-Kavach AppSec SDK addresses these challenges by monitoring the secure posture of the device at runtime, identifying potential threats, and enforcing robust security.
- This enables developers to seamlessly integrate the SDK to deliver secure and compliant mobile applications efficiently, without impacting development timelines.



APPSEC SDK

**25+**
SECURITY CHECKS

1. OS & SYSTEM INTEGRITY CHECKS

6
CHECKS

- 1 Root Detection
- 2 Emulator Detection
- 3 Device Attestation
- 4 Developer Options Detection
- 5 Play Protect Status Detection
- 6 Device Time Tamper Detection



2. NETWORK & CONNECTIVITY SECURITY

6
CHECKS

- 1 Secure WiFi Detection
- 2 Proxy Detection
- 3 VPN Detection
- 4 User Installed Certificates Detection
- 5 Hotspot Status Detection
- 6 NFC Status Detection



3. APPLICATION, RUNTIME & MALWARE PROTECTION (RASP)

7
CHECKS

- 1 Debug Status Detection
- 2 Dynamic Instrumentation Tools Detection
- 3 Overlay Detection
- 4 Screen Captures Prevention
- 5 Installation Source Detection
- 6 Accessibility Service Detection
- 7 Remote Applications Detection



4. DEVICE SECURITY & ACCESS CONTROL

6
CHECKS

- 1 Device Encryption Status Detection
- 2 Device Lock Screen Detection
- 3 Device Lock Screen Strength Detection
- 4 Biometric Enrollment Detection
- 5 Show Password Status Detection
- 6 USB Debugging Detection

APPSEC SDK
25+ SECURITY
CHECKS

5. HARDWARE & DEVICE TELEMETRY



- 1 RAM & ROM Information of the Device



RAM



ROM

Additional Device Telemetry

- CPU Architecture
- OS Version
- Build Information
- Device Model
- Bootloader Status
- And more...

Multi-Layer
ProtectionLightweight
& EfficientReal-time Threat
DetectionEasy Integration
(Plug & Play)Highly Configurable
& ScalableEnterprise
ReadySECURE. DETECT. PROTECT. | POWER YOUR APP SECURITY WITH **APPSEC SDK**



iOS SECURITY FEATURES

10+
SECURITY CHECKS

1. OS & SYSTEM INTEGRITY CHECKS



- 1 Jailbreak Detection**
Detects if the device is jailbroken or compromised.
- 2 Simulator Detection**
Identifies if the app is running on a simulator.
- 3 Lockdown Mode Detection**
Checks if Lockdown Mode is enabled for enhanced security.



3. NETWORK & CONNECTIVITY SECURITY



- 1 VPN and Proxy Detection**
Detects if the device is connected through VPN or Proxy to identify potential security risks.



2. APPLICATION & RUNTIME PROTECTION



- 1 Instrumentation Tools Detection**
Detects presence of instrumentation tools that can manipulate the app.
- 2 Debugger Detection**
Detects if a debugger is attached to the app.
- 3 Screen Captures Prevention**
Prevents unauthorized screenshots and screen recording.



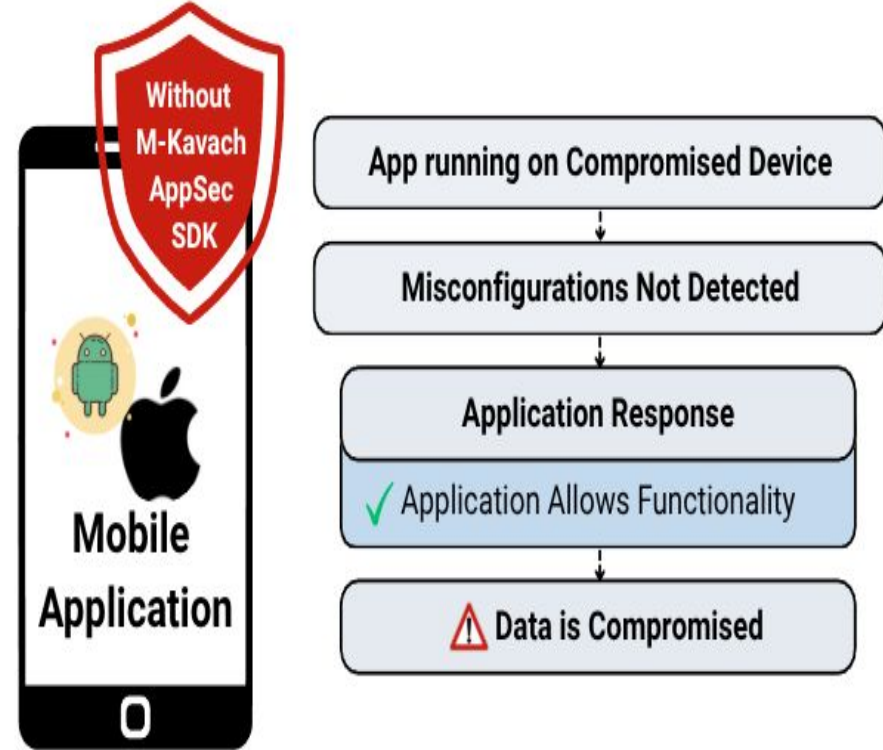
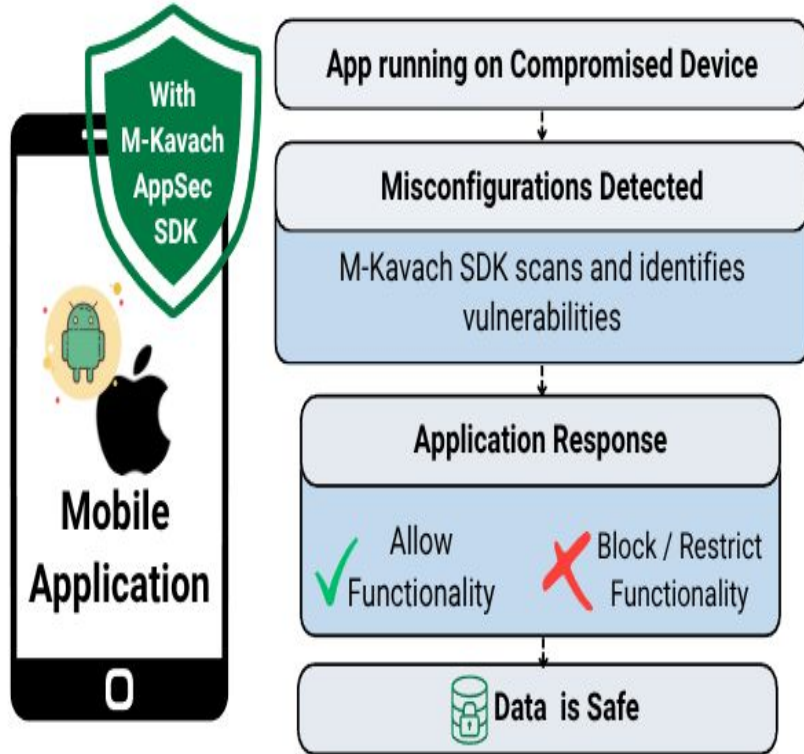
4. DEVICE SECURITY & USER AUTHENTICATION



- 1 Device Encryption Status Detection**
Checks if device encryption is enabled.
- 2 Biometric Registered/ Enrolled Detection**
Verifies if biometric authentication is registered or enrolled.
- 3 Device Passcode Detection**
Checks if a secure device passcode is set for device protection.

Advanced Protection
for iOS DevicesLightweight
& EfficientReal-time
Threat DetectionEasy Integration
(Plug & Play)Low Overhead
High PerformanceEnterprise
Ready

Secure Your Application with M-Kavach AppSec SDK



MAJOR ACHIEVEMENTS



Deployed in the
Digital Census Mobile Application



Successfully running on
33+ lakh (3.3M+) devices

33+
LAKH+



Supported on both
Android & iOS platforms



Proven stability and scalability
at large scale



M-Kavach 2

Secure Mobile Application



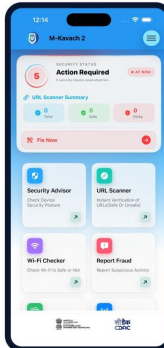
M-KAVACH 2

M-Kavach 2 is a comprehensive mobile device security solution addressing emerging threats related to mobile devices. The major emphasis is on advising the users against security misconfigurations, detection of hidden apps and scanning the device for potentially risky apps installed on the user's mobile device.

Salient Features

ANDROID

- Security Advisor :** Checks your phone's security settings and suggests easy fixes to stay safer.
- Hidden Applications :** Identifies the existence of any hidden apps on the user's device
- Threat Analyzer :** Detects potentially risky apps on the user's device.
- App Statistics :** Notify the users regarding the apps not updated for longer durations
- Adware Scan :** Notify the users about the adware present on the device.
- App Verifier :** Confirms the authenticity and developer ownership through a Blockchain enabled solution
- URL Scanner :** Detects safe or unsafe URLs by scanning web links.
- Report Fraud :** Enables reporting of scam links, fraudulent activities, and safety concerns.



Android

IOS

- Security Advisor :** Checks your phone's security settings and suggests easy fixes to stay safer.
- URL Scanner :** Detects safe or unsafe URLs by scanning web links.
- Wi-Fi Checker :** Checks if your connected Wi-Fi network is secure and alerts you to risks.
- Report Fraud :** Enables reporting of scam links, fraudulent activities, and safety concerns.



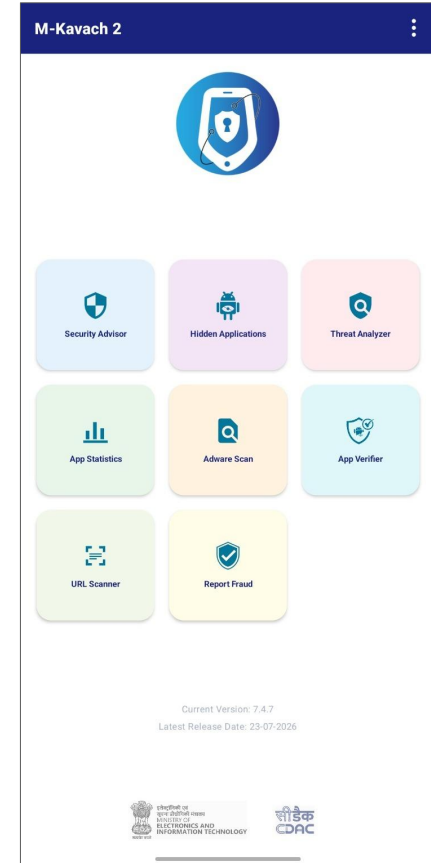
IOS

Contact Details : mkavach@cdac.in [+91 90142 02885](tel:+919014202885)

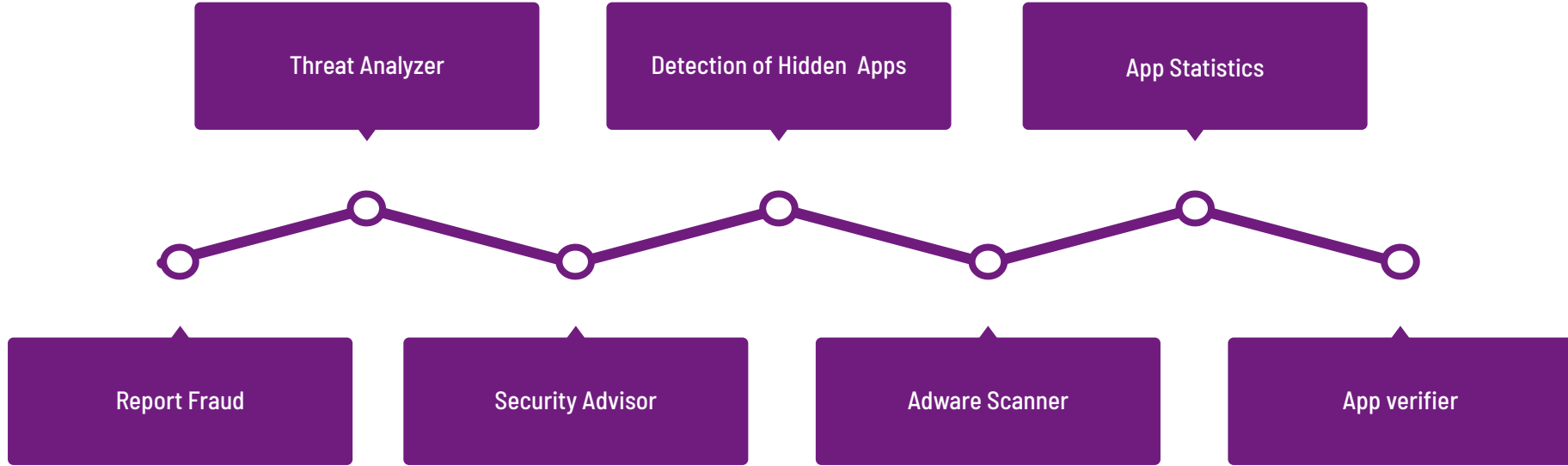
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CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

M-Kavach 2

M-Kavach 2 is a comprehensive mobile device security solution addressing emerging threats related to Android based mobile devices. The major emphasis is on advising the users against security misconfigurations, detection of hidden apps and scanning the device for potential malicious apps installed on the user's mobile device.



Salient Features



Major Achievements

- Published in the Google Play Store, MSeva App store & Cyber Swachhta Kendra with a total of 5.6M+ downloads.
- Headquarters, Integrated Defence Staff, Ministry of Defence has issued an advisory to all defence agencies, recommending the use of the M-Kavach 2.0 app
- Integrated with Secure Application for Internet (SAI) app of Indian Army and MSeva Master App

App Stores Availability



ios



Google Play Store

<https://play.google.com/store/apps/details?id=org.cdac.updatemkavach>



Scan & Install

Parikshan

Automated Security Analysis of Mobile Apps



PARIKSHAN

Insightful App Security Analysis

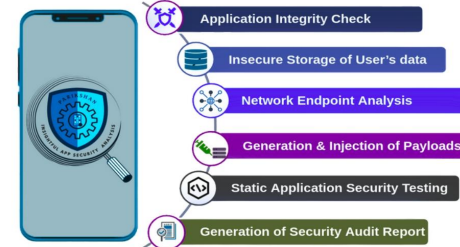
Mobile applications witnessed rampant proliferation from multiple App Stores and these applications handles user's sensitive information and critical device resources. Hence, the security assessment & penetration testing of the mobile applications has become paramount in-order to prevent mis-utilization of data & resources. Apparently, an automation tool performing fast pace security analysis becomes inevitable. Parikshan is an Automation tool mainly focused on performing static and dynamic analysis of mobile applications. The tool has the capabilities to identify the security vulnerabilities and perform penetration testing for few of them. As an outcome, a detailed security audit report is generated containing the information about the identified vulnerabilities which aids to carry out further analysis.

Salient Features:

- Automated Security Analysis, Vulnerability Assessment & Penetration Testing of Mobile Applications.
- Adheres to OWASP Top 10 Mobile Vulnerabilities and executes a total of 24 Static & 12 Dynamic test cases.
- Analysis of device logs and storage to identify leakage of user sensitive data.
- Generation and Injection of payloads during Dynamic Analysis.
- Real time capture of Geo-location information of the network endpoints communicated by the application
- Generation of detailed Audit Report of identified Vulnerabilities

Target Audience :

- Mobile Application Developers
- Mobile Application Security Analysts



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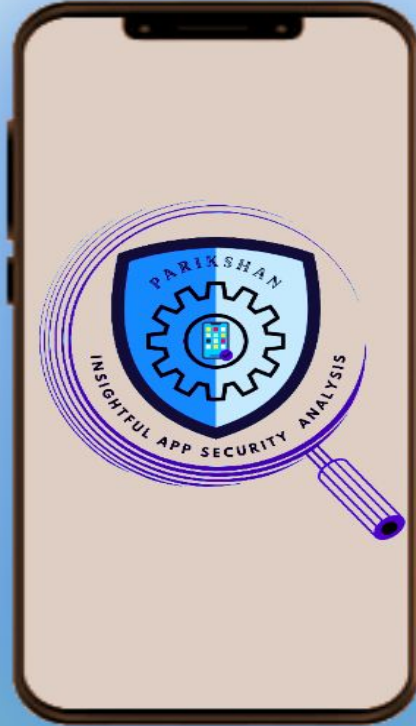
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CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

Parikshan:Automation tool

- ✓ Mainly focused on performing Static and Dynamic analysis of mobile applications.
- ✓ Capable to identify the security vulnerabilities and perform exploitation for few of them.
- ✓ As an outcome, a detailed Security Audit Report is generated.
- ✓ Helps the Security Analysts to carry out further deeper analysis.



Salient Features



Analysis of Source Code



Anti Reversing Defenses



Insecure Storage of User's data



Network Endpoint Analysis



Native API Monitoring



Generation & Injection of Payloads



Runtime Code Injection

THANK

YOU!